A Follow-up Diagnostic Experiment on LongMemEval-S

Yifan Cheng

October 2026

1 Introduction

In my previous LongMemEval-S experiment, I compared history-selection strategies and analyzed their remaining failures. The implementation issue was that the original session identifiers, including identifiers beginning with `answer_`, appeared in the model input. Because these identifiers can indicate evidence-bearing sessions, they create a potential source of *data leakage*.

I revised the code to replace the original identifiers in prompts with pure, sequential session numbers. Original IDs are retained only as evaluation metadata. I then reran BM25 with `gpt-6-luna` at `none` and `medium` reasoning effort on the same 25 questions, using an 8,000-token history budget.

I also added two diagnostic measures. First, *evidence coverage* records whether the selected history contains all, some, or none of the annotated supporting sessions. Second, the *Oracle* condition directly supplies all annotated supporting sessions to the model, without including reference answers or annotation markers in its input. This helps separate evidence-selection failures from failures to interpret available evidence.

2 Corrected BM25 Results and Evidence Coverage

Each of the 25 questions was evaluated once with Luna (none) and once with Luna (medium), for 50 corrected BM25 requests in total. The overall accuracies were shown at table 1. This stability does not establish that session IDs had no effect on individual predictions; the correction primarily removes a methodological confound.

Let G_i be the set of annotated supporting sessions for question i, and let S_i be the sessions selected by BM25. I define per-question evidence recall as

$$R_i = \frac{|G_i \cap S_i|}{|G_i|}, \qquad \overline{R} = \frac{1}{N}\sum_{i=1}^{N} R_i. \tag{1}$$

Coverage is *all* when every annotated supporting session is retrieved, *some* when only a nonempty subset is retrieved, and *none* when no annotated supporting session is retrieved.

There is one important exception to the relationship between complete evidence and correctness: medium achieves 18 correct answers despite obtaining *all* annotated evidence in 19 cases. The additional case, `a2f3aa27`, concerns Instagram followers. The history confirms 1,250 followers and later says the user is *close to* 1,300, whereas the reference answer is 1,300. Medium avoided asserting an exact new count. I marked this answer incorrect under strict benchmark scoring, but it is a plausible case of reference-answer ambiguity rather than a clear reasoning error.

Table 1: Results on the same 25 questions after session-ID sanitization.

Reasoning effort	All	Some	None	Mean recall	Correct	Accuracy
None	19 (76%)	3 (12%)	3 (12%)	0.816	13/25	52%
Medium	19 (76%)	3 (12%)	3 (12%)	0.816	18/25	72%

3 Oracle Diagnosis of the Six Remaining Medium Failures

For the Oracle condition, I increased the configurable input-token guard to 18,000 to accommodate all annotated supporting sessions without truncation. The BM25 history budget remained fixed at 8,000 tokens.

The six BM25 errors with incomplete evidence were evaluated again under the Oracle condition. Under strict benchmark scoring, Oracle corrected 4/6 with medium and 2/6 with none. Table 2 summarizes the observations.

Table 2: Oracle results on six selected BM25 failures. Both efforts received the same annotated supporting sessions.

Question ID	BM25 evidence	Oracle: none	Oracle: medium
`3ba21379`	None	Ford F-150 (**Correct**)	Ford F-150 (**Correct**)
`gpt4_8279ba03`	None	Smoker (**Correct**)	Smoker (**Correct**)
`5a7937c8`	Some	"I don't know" (**Incorrect**)	Three days: Dec. 10, 17, 24 (**Correct**)
`2b8f3739`	None	$345 (**Incorrect**)	$495 (**Correct**)
`bf659f65`	Some	"At least two" (**Incorrect**)	Two albums/EPs (**Incorrect**)
`gpt4_d6585ce8`	Some	Incomplete concert order (**Incorrect**)	Queen concert not placed (**Incorrect**)

3.1 Two medium-effort failures: ambiguity in the reference or evidence

`bf659f65` (album/EP counting). The reference answer is three, but the explicit user statements establish two qualifying actions: downloading Billie Eilish's *Happier Than Ever* and buying The Whiskey Wanderers' *Midnight Sky* EP. A third annotated session says that the user had a Tame Impala vinyl record signed after a show. Possession is clear, but its *purchase or download* is not explicitly stated. Medium answered two, while none answered *"at least two."* The latter is interesting because it acknowledges the lower bound without assuming the unconfirmed third purchase. Both answers are scored wrong against the reference, but neither is an unambiguous reasoning failure.

`gpt4_d6585ce8` (concert ordering). The reference lists five events and places Queen + Adam Lambert last. The annotated conversations indicate the dates of several earlier events, but the Queen statement says the user had *just seen* the concert before the April 15 conversation, without giving its exact date. Medium correctly ordered the other four events and explicitly declined to place Queen precisely; none omitted it from its ordered list. Thus the strict reference was not reproduced, although the original evidence may not support a unique complete chronological ordering. This is a second case where uncertainty should not automatically be labeled a reasoning failure.

3.2 Two additional none-effort failures: calculation and cross-session integration

`2b8f3739` (arithmetic). None extracted the relevant amounts, $225, $150, and $120, but concluded $345. The correct calculation is

$$225 + 150 + 120 = 495. \tag{2}$$

Medium returned $495. This is a clear arithmetic failure under none effort.

`5a7937c8` (multi-session integration). The supporting sessions describe activities on December 10, 17, and 24. None still responded that the total number of days was unknown, whereas medium combined the three sessions and correctly answered three days. This suggests a difference in integrating evidence across sessions, not a retrieval difference in this Oracle comparison.

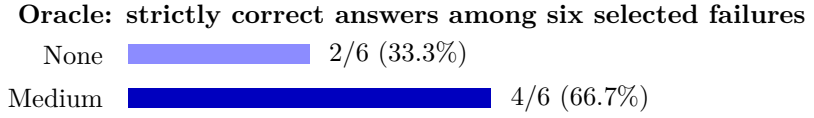


Figure 1: Diagnostic Oracle outcomes on the same six selected questions.

4 Conclusion and Discussion

Within this small 25-question study, medium-effort Luna performed well when *both* relevant evidence was fully retrieved and the answer was sufficiently determined by the text. Compared with none, medium also handled arithmetic and multi-session integration better in the Oracle comparison. The observed differences suggest that additional reasoning effort can improve calculations and evidence combination, while a cautious interpretation may still diverge from an over-specific benchmark answer.

These observations have limitations. Only six BM25 failures were selected for Oracle, and the sample is small. Moreover, in five of those six cases, the Oracle history exceeded the 8,000-token BM25 budget. Oracle therefore serves as an *evidence-sufficiency diagnostic*, not a controlled comparison at equal context length. Providing only annotated sessions also removes distractors.